

Original research

Acute neuromuscular and fatigue responses to the rest-pause method

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Received 29 November 2010; received in revised form 12 August 2011; accepted 19 August 2011

Abstract

Objectives: To compare muscle recruitment, maximal force, and rate of force development changes following different resistance exercise protocols with a constant volume-load.

Design: Within-subjects randomized crossover trial.

Methods: Fourteen ($n = 14$) resistance trained male participants completed three different resistance exercise protocols involving 20 squat repetitions, prescribed at 80% of 1-repetition-maximum. Protocol A consisted of 5 sets of 4 repetitions with 3 min inter-set rest intervals, protocol B was 5 sets of 4 repetitions with 20 s inter-set rest intervals, and the rest-pause method was an initial set to failure with subsequent sets performed with a 20 s inter-set rest interval. Maximal squat isometric force output and rate of force development (RFD) were measured before, immediately upon completion (IP), and 5 min (5P) following each protocol. Muscle activity from 6 different thigh and hip muscles was measured with surface electromyography (EMG) at each time point, and during every squat repetition.

Results: Participants completed the rest-pause method in 2.1 ± 0.4 sets, with a total protocol duration of 103 s compared to 140 s and 780 s for protocols B and A, respectively. All protocols elicited similar decreases ($p < 0.05$) in maximal force and RFD at IP, with full recovery at 5P. Increased motor unit recruitment was observed during the rest-pause method compared to both protocols A and B for all muscles measured ($p < 0.05$).

Conclusions: As a result of the increased EMG during exercise and no greater post-exercise fatigue, it was concluded that the rest-pause method may be an efficacious training method for resistance-trained individuals.

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Keywords: Rest-pause; Training volume; Training intensity; Electromyography; Fatigue

1. Introduction

In the construction of resistance training programs a number of variables may be manipulated in attempts to achieve specific goals. One particular resistance exercise program used in the recreational training environment (particularly bodybuilding) that manipulates the use of repetitions to failure and very short inter-set rest intervals is the rest-pause method. The rest-pause method typically involves prescription of a fixed volume-load (e.g., 20-repetitions of the squat exercise, or 30-repetitions of an arm exercise) where after an initial failure set (typically 10–12 repetitions) subsequent failure sets are performed as required using short (e.g., 20 s) inter-set rest intervals. In previous research, the

rest-pause method has been described as a prolonged pause between individual repetitions within a given working set.¹ A prolonged pause between individual repetitions is not a commonly observed technique in either the resistance training research or clinical practice. There is no research investigating the rest-pause method of current interest (i.e. the performance of a fixed volume-load constituted of repeated failure sets with very short rest periods).

Muscular failure and the associated metabolic and hormonal responses are considered to provide a significant stimulus for strength and neuromuscular adaptations with training.^{2,3} It is thought that the extent of muscular fatigue elicited by performing repetitions to failure determines the efficacy of the training stimulus.⁴ A more fatiguing resistance exercise bout is often attained by increasing the total number of failure sets, thereby increasing the total volume of exercise performed.⁵ The rest-pause method is unique in that

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the prescribed volume-load is collapsed into as short a time period as possible. It is reasonable to believe the rest-pause method will be more fatiguing than protocols composed of the same volume-load but not involving repetitions to failure.

Manipulation of inter-set rest interval has been shown to significantly influence acute hormonal and metabolic responses.⁶ The length of rest interval dictates the recovery that occurs between sets. At 20 s, approximately 50% of adenosine triphosphate and creatine phosphate are restored, whereas at 3 min approximately 85% are restored.⁷ The American College of Sports Medicine recommends resistance trained individuals use a rest interval length of between 2 and 3 min for strength and hypertrophy.⁸ However, there is equivocal evidence for whether or not short or longer rest intervals are more effective for strength and hypertrophy gains.^{9,10} Typically, relative training intensity is reduced from set to set in order to allow matched volume,^{11–13} forced or assisted repetitions are performed in order to ensure prescribed repetitions are achieved,¹⁴ or low relative training intensities (<60% of a 1-RM) are used to ensure that the short duration rest periods do not cause repetition failure and/or that total volume is matched.^{15,16} The rest-pause method is unique in that there is no reduction in the training intensity, or assistance provided, in the attainment of prescribed volume.

The importance of the fatiguing stimulus associated with an acute resistance training bout is associated with two physiological mechanisms for adaptation. First, it is thought that high-intensity fatiguing protocols lead to greater motor unit recruitment compared to non-fatiguing protocols.⁴ This is associated with Hennemann's size principle, which describes the orderly recruitment of progressively larger motor units (and the corresponding muscle fibers) in response to increases in the intensity of the stimulus.¹⁷ Increased motor unit recruitment is thought to be crucial for facilitating maximal strength development,⁶ especially in more advanced resistance-trained individuals.¹⁸ However, the measurement of motor unit recruitment is not often recorded during the working repetitions. Therefore insight into how different resistance exercise protocols influence recruitment is unclear. Second, more fatiguing protocols (typically associated with higher volumes of exercise) are associated with greater muscle protein signaling responses,^{19,20} which may facilitate greater protein accretion in muscle over time.

The purpose of this study was to measure the acute fatigue and motor unit recruitment changes during and after 20 squat repetitions performed using the rest-pause method, and similar volume-loads performed in a non-failure manner using short (20 s) and long (3 min) inter-set rest intervals. The hypotheses of this study were that the rest-pause method would facilitate greater muscular fatigue and increased motor unit recruitment.

2. Methods

Fourteen healthy, resistance-trained males volunteered to participate in this study after providing informed writ-

ten consent. Participant characteristics are as follows (mean $\pm$ SE): age, 25.0 ± 1.7 years; height, 1.80 ± 0.01 m; weight, 85.8 ± 2.3 kg; training experience, 5.5 ± 0.8 years; squat 1-RM, 179.3 ± 8.1 kg. The University Human Participants Ethics Committee approved all procedures used in this investigation. All participants had been performing regular resistance exercise at least 3 days per week for the previous 2 years. Participants reported that they were not taking performance enhancing stimulants at the time of testing and had no musculoskeletal injuries or disorders. Participants were instructed to refrain from any resistance or anaerobic exercise and were required to maintain normal dietary habits in the 24 h preceding the testing sessions. Participants were required to present to testing in a 2-h postprandial state.

Each experimental protocol involved performance of 20 squat repetitions, at an intensity of 80% of 1-RM. All squat repetitions were performed to a depth of 90° of knee flexion. This exercise model (type and intensity) has recently been used to discriminate strength gains between different volumes of prescription after as little as 3-weeks training.⁵ After an initial familiarization session in which participant's squat 1-RM was measured and they were familiarized with the testing protocols, participants were randomly assigned to perform each of the three experimental protocols. Each testing session was separated by 5–7 days, and testing was always performed at the same time of day. The three experimental protocols were: protocol (A) 5 sets $\times$ 4 repetitions, with 3 min inter-set rest intervals; protocol (B) 5 sets $\times$ 4 repetitions, with 20 s inter-set rest intervals; and the rest-pause method in which repetitions were performed to failure followed by a 20 s rest interval. Subsequent sets were performed until 20 repetitions were accrued. Tow-second eccentric and one-second concentric movement phases were used for each repetition to ensure comparable time under tension between protocols. No participant failed during a squat repetition during protocol A or B.

To investigate fatigue changes, squat force output and rate of force development (RFD) were recorded via maximal voluntary isometric squat contractions (MVIC) before (PRE), immediately post (IP), and 5 min post-exercise (SP) each protocol. Electromyography (EMG) data were collected from selected muscles during the MVICs and throughout the protocols in order to examine motor unit recruitment. After careful skin preparation using disposable razors to remove excess hair, fine sandpaper and isopropyl alcohol swabs to reduce skin electrode impedance to below 5 k Ω , pairs of blue dot silver/silver-chloride electrodes (Maxensor, MediMax Global, Australia) with a 1.4 cm diameter and 2 cm center to center distance were applied to the following muscles: erector spinae (ES), gluteus maximus (GM), biceps femoris (BF), rectus femoris (RF), vastus lateralis (VL) and vastus medialis (VM).²¹ Each pair of electrodes was placed according to previous recommendations for ideal anatomical placement.²¹ This included measurement of relevant bony landmarks and other prominences to maintain consistency of placement among participants. Electromyography



Fig. 1. Power rack setup for isometric squat test on Kistler force platform.

graphic signals were recorded using telemetric transmitters (Aurion Inc, Italy; common mode rejection ratio of 90 dB at 50 Hz; input impedance 100 M Ω , -6 dB band pass roll off at 10 and 1000 Hz) at 2000 Hz with 16 bit analog to digital conversion. All collected signals were subsequently band pass filtered (between 10 and 500 Hz), then rectified and smoothed using a root mean square (RMS) calculation with a 50 ms sliding window.²² The maximal RMS EMG (mV) for each muscle was recorded during the isometric squat trials at each time point and during each repetition performed during the experimental protocols.

Participants performed a 1-RM barbell squat during the familiarization session as a measure of dynamic lower limb strength according to standard procedures.^{23–25} The highest achieved weight (kg) prior to failure was recorded as the 1-RM. Parallel safety bars were used and a 2-point person spot was provided for all trials. Participants were provided with verbal encouragement throughout 1-RM attempts.

Maximal isometric squat strength was measured PRE, IP, and 5P from the vertical reaction force (Fz) recorded on a Kistler piezoelectric force platform (Kistler Instruments Pty Ltd, Australia). Between 3 and 5 sub-maximal warm-ups were provided at the start of each testing session, with 1 min rest between repeats. Participants were positioned in the 90° isometric squat position, with solid parallel bars resisting movement (Fig. 1) and were instructed to contract “as fast and forcefully as possible” in a vertical direction against the resistance of the parallel bars. Trials with an initial counter-movement (identified by a visual drop in the force trace) were always excluded, and a new trial performed. Two maximal trials were performed at each time point (PRE, IP, 5P), with 1 min rest between trials. Force signals were sampled at 1000 Hz using an analog to digital converter (National Instruments, USA). All recorded force outputs were corrected for the combined body and bar weight of the participants. Force platform signals were subsequently smoothed using a 4th order, digital low pass filter at 20 Hz. The trial with

the maximum force value at each time point was used for data analysis. Squat RFD was calculated as the maximum slope of the force–time ($\Delta\text{force}/\Delta\text{time}$) curve. The onset of contraction was defined as the first time point on the force–time curve that exceeded the pre-load force level by >15 N.

Maximal EMG (mV) was recorded during each repetition of each protocol for all muscles measured. For data analysis, the average amplitudes of repetitions 1–4, 5–8, 9–12, 13–16 and 17–20 were calculated. This allows comparison between equivalent repetitions performed during each protocol. The total number of sets required, and repetitions per set, to complete the rest-pause method were recorded.

Between-session reliability of maximal force output and squat RFD were calculated using PRE testing results recorded prior to each experimental protocol (ICC, r -value). Maximal force (N), squat RFD (N s⁻¹), and muscle activity (mV) during isometric squat trials were analyzed using a protocol (A, B, RP) $\times$ time (PRE, IP, 5P) repeated measures analysis of variance (ANOVA). Muscle activity between sets of repetitions was compared using a protocol (A, B, RP) $\times$ set (1–5) repeated measures ANOVA. When main effects were observed in the ANOVA, Tukey’s post hoc test was used to identify differences. The significance level was set at $p < 0.05$. Data is presented as mean $\pm$ SE.

3. Results

Between-session reliability (based on comparison of PRE test results) was $r = 0.87$ for maximum force, $r = 0.80$ for squat RFD, and ranged between $r = 0.60$ and 0.80 for maximal EMG activity across the different muscles. Participants required 2.1 ± 0.4 sets to complete the 20 repetitions of the rest-pause method. Average repetitions per set were 14.9 ± 1.4 , 5.9 ± 0.6 , 3.3 ± 0.4 , 2.7 ± 0.2 , and 2, for the 1st, 2nd, 3rd, 4th, and 5th sets, respectively. Only 1 participant required a 5th set to complete the rest-pause method. Based on the 20 s inter-set rest interval, the rest-pause method required a total rest period of 42.9 ± 7.8 s to complete the 20 repetitions, compared to 80 s total rest during protocol B, and 720 s total rest during protocol A.

No significant time $\times$ protocol effects were observed for maximal isometric force and squat RFD. A significant main time effect was observed for both maximal force output and squat RFD ($p < 0.05$). Maximal force was reduced by 8.2% and squat RFD was reduced by 11.5% at IP compared to PRE for the pooled results. Maximal force output and squat RFD were not different from PRE at 5P (Table 1). A significant protocol $\times$ time effect was observed for changes in VM maximal EMG ($p < 0.05$). At IP, as compared to PRE, protocol A and B VM activity decreased by 10.6% and 11.9%, respectively. Following the rest-pause method, VM activity was increased by 8.4% at IP, as compared to PRE. Muscle activity at 5P was not different between protocols.

Table 1

Maximal isometric force output (N) and squat rate of force development (RFD) ($N s^{-1}$) measured before (PRE), immediately post (IP), and 5-min post (5P) each exercise protocol ($n = 14$). Data are mean $\pm$ SE.

	Protocol A	Protocol B	Rest-pause method
Maximal isometric force output (N)			
PRE	1203.5 $\pm$ 99.8	1144.6 $\pm$ 84.6	1198 $\pm$ 75.2
IP	1105.6 $\pm$ 94.6*	1054.4 $\pm$ 89.9*	1095.3 $\pm$ 64.4*
5P	1219.9 $\pm$ 89.3	1146.6 $\pm$ 99.4	1169.1 $\pm$ 81.6
Maximal squat RFD ($N s^{-1}$)			
PRE	4204.3 $\pm$ 584.6	4517 $\pm$ 723.9	4562 $\pm$ 571.5
IP	4057.2 $\pm$ 504.4*	3668.2 $\pm$ 537.9*	4038.2 $\pm$ 676.3*
5P	4667.6 $\pm$ 583.8	4260.3 $\pm$ 793.9	4494.9 $\pm$ 723.5

* $p < 0.05$ as compared to PRE.

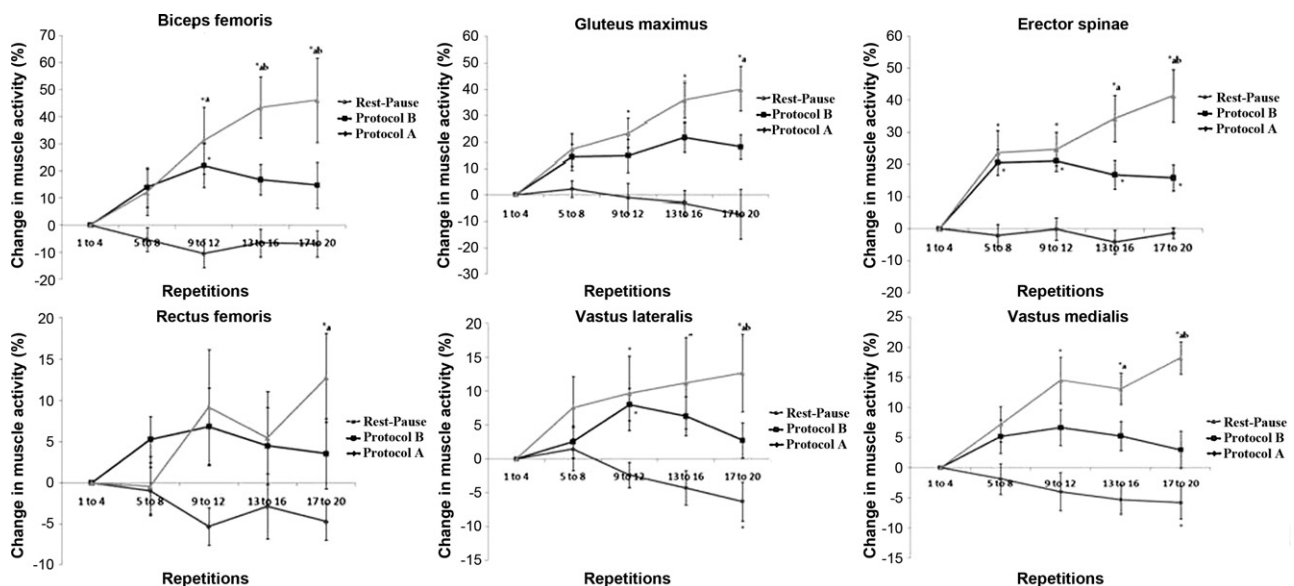
Significant protocol $\times$ set effects were observed for all muscles (Fig. 2; $p < 0.001$). BF muscle activity increased by $46.1 \pm 15.6\%$ during the rest-pause method. This was significantly different to protocol A ($-6.9 \pm 4.9\%$; $p < 0.001$) and protocol B ($14.7 \pm 8.5\%$; $p = 0.01$). ES muscle activity increased by $41.4 \pm 8.2\%$ during the rest-pause method. This was significantly different to protocol B ($15.9 \pm 4.1\%$; $p = 0.005$), and protocol A ($-1.4 \pm 1.7\%$; $p < 0.001$). GM muscle activity increased by $40.3 \pm 8.4\%$ during the rest-pause method. This was significantly different to protocol A ($-7.4 \pm 9.4\%$; $p < 0.001$), but was not different to protocol B ($18.2 \pm 4.8\%$). RF muscle activity increased by $12.8 \pm 5.4\%$ during the rest-pause method. This was significantly different to protocol A ($-4.7 \pm 2.2\%$; $p = 0.014$), but not different to protocol B ($3.5 \pm 4.3\%$). VL muscle activity increased by $12.7 \pm 5.7\%$ during the rest-pause method. This was significantly different to protocol A ($-6.3 \pm 2.9\%$; $p < 0.001$) and protocol B ($2.8 \pm 2.6\%$; $p = 0.001$). VM muscle activity increased by $18.3 \pm 2.7\%$ during the rest-pause method. This was significantly different to protocol A ($-5.8 \pm 2.8\%$; $p < 0.001$) and protocol B ($3.0 \pm 3.1\%$; $p = 0.001$). There were

no differences between protocols A and B with respect to EMG activity recorded for any of the monitored muscles during exercise.

4. Discussion

Some support was found for our hypothesis, with greater motor unit recruitment observed during the rest-pause method as compared to protocols A and B. Surprisingly, the rest-pause method was not accompanied by greater post-exercise reductions in squat force output or RFD. There are two particular findings that require discussion: the absence of between protocol differences in fatigue and the increased motor unit recruitment observed during the rest-pause method.

Similar reductions in RFD and force output were observed immediately following each protocol. Furthermore, similar recovery to pre-exercise levels was observed 5 min post cessation of each protocol. This suggests that either the total volume performed was insufficient to provoke



* is $p < 0.05$ from first set, a is $p < 0.05$ from protocol A, b is $p < 0.05$ from protocol B.

Fig. 2. Change in muscle activity (% change) from the 1st set (repetition 1–4) for the three experimental protocols. Data are mean $\pm$ SE.

between-protocol differences, the rest-pause method was less fatiguing than anticipated, multiple sets regardless of how performed are more fatiguing than expected, or a combination of these explanations. These results suggest that the rest-pause method may be used without concern for performance impairment later in a training session. Similar reductions in maximal force output despite manipulation of independent training variables have been observed for the mode of contraction,²⁶ load,²⁷ muscle group,²⁸ and volume.²⁹

The results of the current study showed a 10.6 and 11.9% decrease in maximal VM EMG following protocols A and B respectively. Conversely, an 8.4% increase in maximal VM EMG was observed immediately following the rest-pause method. Decreases in maximal EMG concomitant to reduced force output have been observed.^{27,29} Changes in EMG concomitant to decreased force output are sometimes interpreted to reflect central activation changes.³⁰ This conclusion may not be appropriate for the current study, considering that no EMG recordings for other muscles changed following any protocol. It may be likely that local muscle metabolic changes, particularly for VM, have mediated the relationship between central drive and peripheral recruitment for the different protocols. Future research should employ more appropriate measures of central activation change, such as the interpolated twitch technique and/or transcranial magnetic stimulation.

As hypothesized, the rest-pause method elicited the greatest increases in motor unit recruitment despite no difference in the observed level of muscular fatigue. The greater motor unit recruitment observed during both short rest protocols (protocol B and rest-pause) is associated with the increased task demand, and probably increased local metabolic demand. Increased motor unit recruitment appears to be a necessary nervous system adaptation in response to the intensity and repetition demand of the exercise bout. The findings of this study suggest that heavy loading and short inter-set rest intervals alone (protocol B) are not sufficient to maximize motor unit recruitment during an acute session. Performance of repetitions to failure is required to elicit the greatest increase in motor unit recruitment. Furthermore, decreases in recruitment were observed in later sets during protocol A and are likely attributed to reduced voluntary drive to the musculature.

It must be noted that the results of this study are only externally valid to the squat exercise performed to 90° knee flexion. This is important considering that participants were capable of performing 15 repetitions of the initial set during the rest-pause method. Furthermore, the ecological validity of protocols A and B may be questioned owing to the low number of repetitions per set (four). Protocols A and B were designed from recent observations that the lowest number of repetitions performed in an initial failure set using the same relative intensity and exercise was five.⁵ This repetition number maintained the internal validity of the non-failure protocols. While protocol B is sometimes employed as a variation on the rest-pause method, a 3-min rest interval is usually

observed when higher repetition failure based training models are utilized.^{5,8}

Although it could be argued that given the similarity in observed fatigue at both IP and 5P there are advantages to protocols A and B (e.g., less demanding and therefore more likely to be complied with), the current investigators propose that owing to the increased motor unit recruitment, the rest-pause method holds the greatest promise for a training stimulus. Due to the acute nature of the present study, comment on chronic implications is speculative.

5. Conclusion

The rest-pause method is a time efficient and potentially efficacious training modality that facilitates increased motor unit recruitment compared to non-failure prescription methods. Prescribed with an appropriate number of repetitions (e.g., 20 squats), acute decreases in maximal force and squat RFD were no more profound using the rest-pause method.

Practical implications

- The rest-pause method elicited the greatest increases in motor unit recruitment for all muscle measured during the squat exercise.
- The rest-pause method was no more fatiguing than prescription schemes that did not include failure based repetitions.
- The rest-pause method is a potentially efficacious training scheme, that when performed at high intensity (i.e. 80% 1-RM), should only be recommended for advanced lifters.
- A basic prescription model of rest-pause training is a two-way body-part split program (chest/shoulders/back, arms/legs), alternated between sessions, and performed three times per week.

Acknowledgement

No financial support was received for this study.

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